

SOURCE

Seoul Living Migration (Saenghwal Idong), administrative-dong product  
dong x weekday x arrival hour x sex x age band x trip type x volume

202001–202607 · 79 of 79 months complete · 424 dong  
1,896 parquet files · 10,186,891,962 rows

WHAT THE PRODUCT DOES NOT RECORD

- no date column: a month is one row per weekday, not per day
- the volume column is the within-month sum over like weekdays (hap), not a daily mean
- cells below 3 are masked, 26% of cells overall, and the loss is not spread evenly:  
20-24 to 40-44 lose 11%-70% of their cells, 45-64 lose under 0.13%, and 0-19 and 65+ lose exactly none
- expansion weights vary with age and are not recorded in the data
- the code space is frozen at 2020 while the districts have moved



EXTERNAL ANCHORS — what closes each gap

official district file

measures the hidden volume:  
2.27 per masked cell, 11.1% of  
volume, not the 1.5 midpoint

contact survey

pins the transmission  
parameter; it does not  
identify it

resident registration

denominators, and the coverage  
profile the data does not  
record



ESTIMATOR

co-arrival in the same dong, the same hour, the same weekday

- age x age matrix A, at three spatial scales (dong 424 / district 25 / city 1)
- excess over the proportionate-mixing null: assortativity, normalised MI, rank

THREE CHECKS, all green before any number is quoted

citation audit · independent second implementation 436/436 · bit-for-bit determinism



THREE CLAIMS

1 the instrument responds

the co-arrival matrix tracks  
the Korean school year in all  
79 months, and the year schools  
stayed shut is flat. The  
calendar is never supplied to  
it. Of the 17 months that reach  
the survey's floor, every one  
is a term month; 3 of 78  
alternative calendar alignments  
match.

Fig 2, Fig 3(c)

2 and it still reads near zero

the departure from  
proportionate mixing is near  
zero at every spatial scale and  
every age resolution. An  
identity fixes what this  
estimator can read at all, and  
a recovery experiment turns  
that into a bound conditional  
on how much sits below the  
cell.

Fig 3, Fig 4, Fig 5

3 which changes an allocation

across the coverage family the  
ranking an allocation rule  
reads moves, and the SEIR cost  
of reading the wrong one  
depends on  $R_0$ : largest at  
1.2-1.5, and 5 of 20 verdicts  
turn on it.

Fig 6, Fig 7